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MILITARY SPECIFICATION

AIR FORCE-NAVY AERONAUTICAL BULLETIN
FITTINGS, FLARED, CLASSIFICATION OF DEFECTS OF

1. Scope and purpose. This Bulletin, when called out in a specification or in a procurement document, establishes a classification of commonly occurring defects applicable to flared fluid fittings.

2. For each flared fluid fitting, this Bulletin lists defects and, opposite them, an applicable defect number.

3. Classification of defects. Defects are herein considered for classification according to the effect they have on safety and usability. Definition of classes is in accordance with Standard MIL-STD-105. However, minor defects as defined in Standard MIL-STD-105 are further categorized as follows:

Minor A - may have a slight effect on usability.
Minor B - has no effect on usability.

3.1 Applicability of classifications. These classifications shall govern the acceptance inspection of fittings when so indicated in the fitting specification or drawing or in the procurement document, and shall be applied to inspection of fittings manufactured under such specifications or drawings.

3.2 Sampling.

3.2.1 When acceptance inspection for these parts is done by sampling, the sampling plan will be specified in the appropriate procurement document.

3.2.2 Sampling shall be in accordance with Standard MIL-STD-105, and the following AQL's shall apply to the classification of defects noted in paragraph 3:

Major - 1.5
Minor A - 4.0
Minor B - 6.5

3.2.3 AQL's shall be applied by class of defects.

3.2.4 An inspection lot shall be as defined in paragraph entitled "Formation of lots" in Section 5 of Standard MIL-STD-105.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, SA-ALC, Kelly Air Force Base, TX 78241 by using the self-addressed Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by letter.

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4. Sources. Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

*5. DETAILED REQUIREMENTS.

5.1 Classification of defects for design standard MS21936 will be found in MIL-STD-1655.

5.2 MS33656 Flared End Design.

Class	Defect No.	Characteristic
Major	101	Incomplete or missing holes
Minor A	201	Cone or seal surface finish
	202	Cone angle
	203	T Thread - To specification
	204	Concentricity - C & D to threads
	205	F Dia - Undercut diameter
	206	Surface finish - F diameter
	207	H - Squareness of face to threads
	208	Q - Undercut width
Minor B	301	A Dia - I. D.
	302	E - Distance across hex flats
	303	J - Thread or end length
	304	B - Distance from end to thread chamfer
	305	D Dia I. D. of cone
	306	C Dia O. D. of cone
	307	K - Hex chamfer diameter
	308	Imperfect threads
	309	45° - Thread chamfer angle
	310	Break edge 0.005 max rad
	311	W - Surface between A and D dias
	312	15° - Hex chamfer angle
	313	Radius - Thread Chamfer at F dia
	314	Concentricity - F dia to thread

5.3 MS24385 Flared End Design, Precision
Identical to MS33656 except table of attributes do not apply where detail drawings specify 100% inspection.

5.4 MS33657 Bulkhead Flared End Design

Class	Defect No.	Characteristic
Major	101	Incomplete or missing holes
Minor A	201	Cone or seal surface finish
	202	Cone angle
	203	T - Thread to specification
	204	Concentricity - C & D dias to T
	205	N - Center undercut width
	206	Surface finish - F dia

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Class	Defect No.	Characteristic
	207	H - Squareness of face to threads
	208	Continuation of same thread lead
	209	F - Undercut diameter
Minor B	301	A Dia - I. D.
	302	E - Distance across hex flats
	303	K - Distance from shoulder to end
	304	B - Distance from thread chamfer to end
	305	Radius thread chamfer at F & C dias
	306	D Dia I. D. of cone
	307	C Dia O. D. of cone and thread relief
	308	45° - Thread chamfer angle
	309	P - Distance from center U cut to end
	310	Y Dia - Shoulder dia
	311	W - Surface between A and D dias
	312	Break edge 0.005 max rad
	313	Radius - C dia at shoulder
	314	L or 0.031 - Shoulder length
	315	Radius - C dia at shoulder
	316	S dia - Flange dia
	317	R - Flange width
	318	Concentricity - F dia to thread
	319	E - Hex chamfer dia
	320	Q - Back undercut width
	321	15° - Hex chamfer angle

5.5 MS24386 Bulkhead Flared End Design, Precision. Identical to MS33657 except table of attributes do not apply where detail drawings specify 100% inspection.

5.6 MS33658 Fitting End, Hose Connection.

Class	Defect No.	Characteristic
Major	101	Hole missing or incomplete
Minor A	201	Burrs or sharp edges
	202	Combination of dimensions A & C which would cause dim B to be under minimum
Minor B	301	D - Bead dimension
	302	1.500 - Fitting end length
	303	0.093 - Dimension from outside of bead to end
	304	0.250 - Dimension from inside of bead to end
	305	0.015 - Radius, corner break
	306	0.031 - Fillet radius of bead
	307	0.125 - Fillet radius of fitting end

5.7 MS33677 Fitting End, Taper Pipe Thread.

Class	Defect No.	Characteristic
Major	101	Hole missing or incomplete
	102	Thread P meet requirements of MIL-P-7105

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Class	Defect No.	Characteristic
Minor B	301	C - Dia (Style B only)
	302	D - Dia (Style C only)
	303	E - Width across hex (Style C only)
	304	F - Dia (Style C only)
	305	G - Dia Passage
	306	H - Dia
	307	J - Chamfer length
	308	L - Effective thread length
	309	M - Dim (Style B only)
	310	O Dia - (Style B only)
	311	90° - Chamfer passage
	312	45° - Thread chamfer
	313	15° - Hex chamfer (Style C only)

5.8 MS33649 Internal Fitting Boss

Class	Defect No.	Characteristic
Major	101	Incomplete or missing holes
	102	Surface finish - detail A
Minor A	201	B - Depth of full thread
	202	D - O-Ring diameter
	203	N - Squareness K to T
	204	Concentricity - T to D
	205	T Thread - To specification
	206	120°- Countersink angle
	207	Concentricity - A to D
Minor B	301	A - Dia - Countersink
	302	C - Dia - Through hole
	303	E - Land depth
	304	J - Tap drill depth
	305	0.040/0.010 Radius - Intersection of counterbore and countersink

5.9 MS20756 Flange.

Class	Defect No.	Characteristic
Major	101	Incomplete or missing holes
Minor A	201	Holes equally spaced (Size -40 thru -112)
	202	G - 12 through 32
	203	L - 40 through 112
	204	B - Counterbore depth
	205	Counterbore and flange parallel within 0.005 TIR
	206	C and D dia concentric within 0.010 TIR
	207	Burrs and sharp edges

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Class	Defect No.	Characteristic
Minor B	301	A - Thickness
	302	C - Dia I. D.
	303	D - Dia I. D.
	304	E - Distance across flats
	305	H - Hole diameter
	306	J - Radius
	307	K - Dia O. D.
	308	L - Hole location
	309	Surface finish across flats
	310	Counterbore fillet radius 0.010 max
	311	G/2 sizes - 12 thru 32
	312	E/2 sizes - 12 thru 32
	313	P Hole location - 56 thru 112
	314	M - Thickness
	315	N - Angle
	316	Identification
	317	150√ finish on face
	318	100√ finish across flats
	319	Finish per specification

5.10 MS20757 Connector and MS20760 Adapter

Class	Defect No.	Characteristic
Major	101	Incomplete or missing holes
	102	Cast parts pressure test
Minor A	201	G - Hole location sizes 12 thru 32
	202	L - Hole equally spaced sizes 40 and 48
	203	Burrs and sharp edges
	204	Sealing surfaces 63√ at c dia
Minor B	301	A - Dia I. D.
	302	B - Dia O. D.
	303	C - Dia at sealing surface
	304	D - Length
	305	E - Distance across flats
	306	F - Width of flange
	307	H - Hole Dia
	308	J - Corner radius
	309	K - O. D. option of Mfg.
	310	L - Hole location
	311	N - 45° Chamfer
	312	P - Undercut width (MS20760)
	313	T - I. D. Cast parts only (MS20757)
	314	V - Wall thickness
	315	G/2 - ℄ Sizes 12 thru 32
	316	E/2 - ℄ Sizes 12 thru 32
	317	0.125 - Radius
	318	0.015 - Radius
	319	Micro finish other than sealing area 125√
	320	Identification

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Class	Defect No.	Characteristic
	321	Finish per specification all other characteristics apply per MS33656 and MS33658

5.11 MS20819 Sleeve

Class	Defect No.	Characteristic
Minor A	201	B - Dia hole thru
	202	C - Dia O. D.
	203	G - Dia O. D. of flared sleeve
	204	Dia B, C, F, and G shall be concentric within 0.005 TIR
	205	90° - Corner to be sharp
Minor B	301	F - Dia O. D. of 33° angle
	302	N - Dia Seat relief angle sizes 2 thru 6 aluminum sleeves
	303	J - Dim shoulder
	304	33° - Sealing angle
	305	M - Dim overall length
	306	1° - Taper angle O. D.
	307	R - Radius seat angle
	308	18° - Sizes 2 thru 6 aluminum sleeves
	309	Break edges 45° at shoulder
	310	Burrs and sharp edges except at the 90° corner. All other characteristics apply within the MS20819 drawing
	311	Identification
	312	Finish per specification

5.12 Shapes - Elbows and Tees having Pipe to Tube Ports MS20822, MS20823, MS20825, MS20826.

Class	Defect No.	Characteristic
Major	101	Incomplete or missing holes
	102	Internal burrs
Minor B	301	V Dim - Across wrench flats
	302	Leg lengths to specification
	303	Drill depth to specification
	304	W - Dim wrench pad width
	305	X - Dim wrench pad width
	306	Fillet / corner radius forging O. D.
	307	90° - Leg angle
	308	Sharp edges and hanging burrs
	309	Identification
	310	Finish per specification

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5.13 MS20913 Plug, Pipe

Class	Defect No.	Characteristic
Major	101	Pipe thread MS33677
	102	Sharp edges and hanging burrs
Minor A	201	Incomplete or missing lockwire hole
Minor B	301	A - Width across flats
	302	B - Height of square
	303	C - Location of lockwire hole
	304	E - Drill depth sizes 3 thru 20
	305	K - Drill Dia sizes 3 thru 20
	306	L - Thread length
	307	M - Radius corner
	308	N - Corner break radius optional
	309	A/2 - C_L Center line, drive dimension
	310	Countersink lockwire hole
	311	Identification
	312	Finish per specification

5.14 MS21932 Single Port Cluster Bolt End Design

Class	Defect No.	Characteristics
Major	101	M - Dia sealing lands
	102	16 RHR Surface finish
	103	0.0005 TIR concentricity - sealing lands
Minor B	301	A - Fitting end length
	302	B - Length over lands
	303	C - Length to front land
	304	D - Cross hole position
	305	E - Length to back land
	306	F - Land chamfer length
	307	K Dia - Cross hole diameter
	308	H Dia - Front diameter
	309	J Dia - Front undercut diameter
	310	L Dia - Center relief diameter
	311	N Dia - Back undercut diameter
	312	45° - Chamfer front and back of thread
	313	0.135/0.090 - Front undercut width
	314	0.056/0.021 Rad - Front undercut radii
	315	0.094/0.031 Rad - Land chamfer radii
	316	30° - Land chamfer
	317	0.026 Rad - Back undercut radius
	318	0.052 - Back undercut width
	319	125 RHR Surface finish-outside machined surfaces
	320	250 RHR Surface finish-inside machined surfaces
	321	Concentricity - L & M to thread T within 0.005

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5.15 MS21934 Double Port Cluster Bolt End Design

Class	Defect No.	Characteristics
Major	101	M - Dia sealing lands
	102	16 RHR - Surface finish - sealing lands
	103	0.0005 TIR concentricity - sealing lands
Minor B	301	A - Fitting end length
	302	B - Length over lands
	303	C - Length to front land
	304	D - Length to front cross hole
	305	E - Length to front of center land
	306	F - Length to back of center land
	307	G - Length to back cross hole
	308	H - Length to back land
	309	J - Front undercut diameter
	310	K Dia - Cross hole dias
	311	L Dia - Relief dias
	312	N Dia - Back undercut dias
	313	P Dia - Front diameter
	314	R - Land chamfer width
	315	45° - Thread chamfer
	316	0.135/0.090 - Front undercut width
	317	0.056/0.025 Rad - Front undercut radii
	318	0.094/0.031 Rad - Land chamfer radii
	319	30° - Land chamfers
	320	0.026 Rad - Back undercut radius
	321	0.052 Rad - Back undercut width
	322	125 RHR Surface finish outside machined surfaces
	323	250 RHR Surface finish inside machined surfaces
	324	Concentricity L & M dia to thread T

5.16 Cluster Fitting Bodies; One, Two and Three Way, Internal Boss and MS33656 Port - MS21946, MS21953, MS21954, MS21955, MS21956, MS21958, MS21959 and MS21960.

Class	Defect No.	Characteristic
Major	101	Incomplete or missing holes
	102	Internal burrs
Minor B	301	Leg length
	302	Leg position to side of body
	303	Leg turn length
	304	Chamfer angle at body and leg
	305	Radius at body and leg
	306	Chamfer dia at body and leg
	307	Angle between legs
	308	Surface finish machined surfaces
	309	Break edges and remove burrs
	310	Marking per specification
	311	Finish per specification

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5.17 MS21947 Single Port Cluster Bolt End Design - Flared.

Class	Defect No.	Characteristic
Major	101	K Dia - Sealing lands
	102	16 RHR Surface finish
	103	0.0005 TIR concentricity - sealing lands
Minor B	301	A - Fitting end length
	302	B - Length over lands
	303	C - Length to front land
	304	D - Cross hole position
	305	E - Length to back land
	306	F - Land chamfer length
	307	G Dia - Cross hole diameter
	308	H Dia - Front undercut diameter
	309	J Dia - Center relief diameter
	310	K Dia - Back undercut diameter
	311	45° - Chamfer back of thread
	312	0.135/0.090 - Front undercut width
	313	0.056/0.021 Rad - Front undercut radii
	314	0.094/0.031 Rad - Land chamfer radii
	315	30° - Land chamfer
	316	0.026 Rad - Back undercut radius
	317	0.052 - Back undercut width
	318	125 RHR Surface finish-outside machined surfaces
	319	250 RHR Surface finish-inside machine surfaces
	320	Concentricity - J & K dias and thread T within 0.005
	321	Threads - Per specification

5.18 Cluster Bolts, Single and Double Port MS21949, MS21950 and MS21951.

Class	Defect No.	Characteristic
Major	101	Incomplete or missing holes
	102	Internal burrs
Minor B	301	B Dia - Overall length
	302	E Dia - Drill depth past cross hole
	303	C Dia - Drill depth, flared end
	304	F Dia - Drill hole dia
	305	H - Hex dimension
	306	Hex chamfer angle
	307	Marking per specification
	308	Finish per specification

5.19 MS24387 Bolt - Flared Tube Fitting Precision.

Class	Defect No.	Characteristic
Major	101	Incomplete and missing holes
	102	Internal Burrs
	103	Serrations to size and form
	104	100% inspection of cone surface condition

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Class	Defect No.	Characteristic
Minor A	201	Concentricity T,A,F, and N to be concentric within 0.005 TIR
	202	A Dia - Shank O. D.
	203	C Dim - Thread length shank of fitting including two imperfect threads
Minor B	301	B Dim - Drilled hole thru shank
	302	D Dim - Length of shank
	303	E Dim - Overall length
	304	F Dia - Serrations O. D.
	305	G Dim - Distance across hex flats
	306	L Dia - Thread chamfer
	307	M Dim - Hole dia thru shank
	308	N Dia - Hole dia thru fitting
	309	45° - Thread chamfer
	310	15° - Hex face chamfer optional
	311	Sharp edges and hanging burrs
	312	Identification
	313	Finish to specification

5.20 Tees, Precision Type MS24388, MS24389, MS24390.

Class	Defect No.	Characteristic
Major	101	Incomplete or missing holes
	102	Internal burrs
Minor B	301	Leg length short flared end
	302	Leg length bulkhead flared end
	303	Leg length boss end
	304	Distance across wrench pad
	305	Angle between legs
	306	Fillet radius
	307	Wrench pad area
	308	Sharp edges and hanging burrs
	309	Identification
	310	Finish to specification
	311	1/8 - Flange thickness when Z dia greater than V

5.21 MS24391, Plug, Bleeder, Precision Type.

Class	Defect No.	Characteristic
Major	101	1/16 Dia - Bleed hole, incomplete or missing
Minor B	301	A Dia - Drill diameter
	302	B Dia - Base of thread chamfer
	303	C - Drill depth
	304	D - Overall length
	305	E Dia - Distance across hex flats
	306	G - Wall thickness (Minimum)
	307	H - Bleeder hole location

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Class	Defect No.	Characteristic
	308	K - Length end to hex face
	309	L Dia - Head recess diameter
	310	0.073 - Wire hole location (L coded parts)
	311	No. 50 drill dia - L/W hole dia (L coded parts)
	312	0.125 - Recess depth
	313	15° to dim E - Hex chamfer
	314	1/32 Rad - Corner undercut to hex face
	315	1/16 drill - Bleeder hole dia
	316	30° chamfer - Back of thread
	317	45° - Thread chamfer
	318	0.005 Concentricity A, F, & T
	319	Machining to specification
	320	Finish to specification
	321	Thread T to specification

5.22 Nipples, Tube, Reducer and Plug Precision Type MS24392, MS24393, MS24399 and MS24404.

Class	Defect No.	Characteristic
Major	101	Incomplete or missing holes
	102	Internal Burrs
Minor A	201	Conical surface and thread to be concentric within 0.005 TIR
Minor B	301	Distance across hex flats
	302	Overall length
	303	S Dim - Depth of drilled hole (MS24399)
	304	A Dim - Depth of drilled hole (MS24404)
	305	Thread relief 1-1/2 threads (MS24393)
	306	Sharp edges and hanging burrs
	307	Identification
	308	Finish per specification

5.23 Elbows and Tee, Bulkhead, Precision Type MS24390, MS24394, MS24395, MS24396.

Class	Defect No.	Characteristic
Major	101	Incomplete or missing holes
	102	Internal Burrs
Minor B	301	Leg length short flared end
	302	Leg length bulkhead flared end
	303	Drill depth short flared end
	304	Drill depth bulkhead flared end
	305	Distance across wrench pad
	306	Angle between legs
	307	Fillet radius
	308	Wrench pad area
	309	Sharp edges and hanging burrs
	310	Identification
	311	Finish to specification

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5.24 MS24397 Reducer, Boss Precision Type.

Class	Defect No.	Characteristic
Major	101	Incomplete or missing holes
	102	Internal Burrs
Minor A	201	K - Thread or end length
	202	L - Undercut width
	203	N Dia - Undercut thread relief
	204	S Pitch and T Pitch to be concentric within 0.010 TIR
	205	N, H, and T Pitch to be concentric within 0.005 TIR
	206	O. D. Threads in accordance with MIL-S-8879
Minor B	301	C - Lockwire location
	302	G Dia - Distance across hex flats
	303	H Dia - Hole I. D.
	304	M - Overall length
	305	P Thread chamfer
	306	1/8" Lockwire location from hex corners
	307	0.065 - 0.075 Lockwire hole diameter 3 places
	308	1/32" Fillet radius
	309	45° - Thread chamfer
	310	30° - Thread chamfer
	311	Sharp edges and hanging burrs
	312	Identification
	313	Finish per specification

5.25 MS24398 Adapter, Straight, Tube to Boss Precision Type

Class	Defect No.	Characteristic
Major	101	Incomplete or missing holes
	102	Internal Burrs
Minor B	301	E Dim - Overall length
	302	C Dim - Drill depth
	303	D Dim - Distance across hex flats
	304	1/32 Radius at hex face and thread relief
	305	15° to D Dimension hex corners
	306	Identification
	307	Finish per specification
	308	Sharp edges and hanging burrs

5.26 Shapes; Elbows, Tees, and Crosses Tube, Precision Type MS24401, MS24402, MS24403.

Class	Defect No.	Characteristic
Major	101	Incomplete or missing holes
	102	Internal Burrs

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Class	Defect No.	Characteristic
Minor B	301	V Dim - Distance across wrench pad
	302	Q Dim - Tube leg length
	303	Q Depth - Drill depth except for MS24403 to be drilled thru
	304	Fillet corner radius forging O. D.
	305	W Dim - Height and width wrench pad area
	306	X Dim - Width wrench pad area (MS24401)
	307	90° - Leg angle
	308	Sharp edges and hanging burrs
	309	Identification
	310	Finish per specification

5.27 Bolts - Fluid Passage - AN774 and AN775

Class	Defect No.	Characteristic
Major	101	Burrs on internal surfaces
	102	Incomplete or missing holes
Minor A	201	C - Length of thread T
	202	T - Thread T to size and form required by MIL-S-8879
	203	A Diameter
	204	Serration - Size and Form
Minor B	301	E - Overall length
	302	D - Bolt length
	303	B - Dimension cross hole location
	304	Concentricity thread T, Dia A, Dia F and Dia N
	305	Burrs or sharp edges on external surfaces
	306	M Dia - Cross hole diameter
	307	L Dia - Thread chamfer
	308	Imperfect threads
	309	G Dim - Across threads
	310	K - Depth of flow hole (AN775 Only)
	311	Marking - Per specification
	312	Finish - Per specification

5.28 Connectors, Multiple Fluid Pressure Line AN776, AN777, AN778 and AN779.

Class	Defect No.	Characteristic
Major	101	Incomplete or missing holes
	102	Internal Burrs
Minor A	201	F Dia - Bore thru
	202	Serrations - Size and Form
Minor B	301	O Dia - Distance across wrench pads
	302	E Dia - Undercut
	303	H Dim - Width
	304	H/2 - Centerline
	305	K Dim - Width of undercut
	306	P - Height of wrench pad

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Class	Defect No.	Characteristic
	307	Q - Leg length
	308	Dia F, M, N to be concentric within 0.005 TIR
	309	Faces to be square with bolt hole axis within 0.006 TIR
	310	0.032 Radius - Undercut corner, 4 places
	311	0.062 Fillet corner radius forging O. D.
	312	Sharp edges and hanging burrs
	313	Identification
	314	Finish per specification

5.29 Shapes, Fittings such as Tee, Elbows and crosses Produced from Forgings- AN783, AN784, AN804, AN821, AN824, AN827, AN833, AN837, AN838, AN839.

Class	Defect No.	Characteristic
Major	101	Incomplete or missing holes
	102	Internal burrs
Minor B	301	Leg length, short flared end
	302	Leg length, Bulkhead flareless end
	303	Leg length, boss end
	304	Leg length, hose end
	305	Drill depth, short flared end
	306	Drill depth, bulkhead flared end
	307	Drill depth, boss end
	308	Drill depth, hose end
	309	Distance across wrench pads
	310	Angle between legs
	311	External corner radius of forgings on elbows
	312	Fillet corner radius of forgings on O.D. of legs at intersections
	313	Radius at back of flange on bulkhead end
	314	Perfect thread depth on boss end
	315	O.D. diameter of boss end
	316	Flange thickness at boss end
	317	Break edges and remove burrs
	318	Marking per specification
	319	Finish per specification

5.30 Straight Fittings such as Unions, Reducers, Expanders, and Plugs Produced from Bar Stock such as AN806, AN807, AN815, AN816, AN832, AN894, AN919.

Class	Defect No.	Characteristic
Major	101	Incomplete or missing holes
	102	Internal Burrs
Minor B	301	Overall length
	302	Length, end to hex face bulkhead flared end
	303	Shoulder width, bulkhead flared end
	304	Drill depth, short flared end
	305	Drill depth, bulkhead flared end
	306	Drill depth, pipe end
	307	Drill depth, internal port

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Class	Defect No.	Characteristic
	308	Distance across hex flats
	309	15° - Hex chamfer angle (AN816, AN894, AN919)
	310	Break edges and remove burrs
	311	Marking - Per specification
	312	Finish - Per specification
	313	Threads - Per specification

5.31 AN814 Plug

Class	Defect No.	Characteristic
Major	101	1/16 Dia - Bleed hole, incomplete or missing internal burrs
Minor A	201	F Dia - Undercut diameter
	202	S - Width of F diameter
Minor B	301	A - Drill diameter
	302	B Dia - Base, thread chamfer
	303	C Dia - Drill depth
	304	D - Overall length
	305	E Dia - Distance across hex flats
	306	H - Bleeder hole location
	307	K - Length - end to hex face
	308	L Dia - Head recess diameter
	309	0.073 Wire hole location (L coded parts)
	310	#50 Drill Dia - Lockwire holes (L coded parts)
	311	0.125 - Recess depth
	312	15° to dimension E - hex chamfer
	313	1/32 Rad - Corner undercut to hex face
	314	1/16 Drill - Bleeder hole diameter
	315	30° Chamfer - Back of thread
	316	45° Thread chamfer
	317	0.005 concentricity A, F and T dimensions
	318	Marking - To Specification
	319	Finish - To Specification
	320	Thread T - To Specification

5.32 AN817 Nut Sleeve Coupling.

Class	Defect No.	Characteristic
Minor A	201	Threads per MIL-S-8879
	202	A Diameter hole thru
	203	F diameter minor dia of threads
	204	Q Dim - Perfect thread
	205	Dia A, F and T to be concentric within 0.005 TIR
Minor B	301	33° Angle sealing surface
	302	E Dim - Radius hole thru to sealing surface
	303	B Depth turned length
	304	J Dim - Hex width
	305	15° ± 5° - Hex chamfer
	306	D Dia - O. D. of lip
	307	C Distance across hex flats

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Class	Defect No.	Characteristic
	308	R Radius turned diameter
	309	M Dim - Overall length
	310	L Full thread length
	311	V Dia - Turned dia flats on V dim nut cause for rejection
	312	45° X N dia chamfer at V dia
	313	90° X H Counter sink at thread
	314	45° X U - Corner break
	315	Sharp edges and hanging burrs
	316	Identification per specification
	317	Finish per specification

5.33 AN818 Nut Coupling.

Class	Defect No.	Characteristic
Minor A	201	Threads per MIL-S-8879
	202	A Diameter hole thru
	203	L Depth sleeve stop
	204	S Depth max imperfect thread
	205	Q Depth perfect thread
	206	Diameters A and thread T to be concentric within 0.005 FIR
Minor B	301	B Depth turned length
	302	C Dia - O. D. turned dia typ. Flats on C dia not cause for rejection
	303	90° ± 5° Countersink X H dimension
	304	J Dim - Hex width
	305	M Dim - Overall length
	306	Chamfered edge 45° ± 5° X N dimension
	307	O Dia - Distance across hex flats
	308	R Radius corner at C diameter
	309	U Chamfer 45° ± 5° two places at A dia hole
	310	15° ± 5° two places at hex
	311	Sharp edges and hanging burrs
	312	Identification per specification
	313	Finish - Per specification

5.34 AN846 Elbow

Class	Defect No.	Characteristic
Major	101	Incomplete or missing holes
	102	Internal burrs
Minor B	301	A Dia - Length to flange
	302	B Dia - Front of thread to flange face
	303	C - Threads
	304	D Dia - ℄ threaded leg
	305	E Dia - Flange length
	306	F Dim - Distance across wrench pads
	307	H Rad - Radius behind flange
	308	J Dia - Body diameter
	309	M Dia - ℄ short leg

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Class	Defect No.	Characteristic
	310	R Rad - External corner radius on elbow
	311	S Dia - Flange diameter
	312	U Rad - Fillet corner radius of forgings on O.D. of legs at intersection
	313	Y Dim - Drill depth short leg
	314	Z Dim - Drill depth threaded leg
	315	1/32 Dim - Washer face on hex
	316	45° Chamfer front of thread
	317	Finish - Per specification

5.35 AN848 Elbow and AN849 Connector

Class	Defect No.	Characteristic
Major	101	Incomplete or missing holes
	102	Internal burrs
Minor B	301	C Dim - Threads
	302	M Dim - Overall length (AN849 Only)
	303	M Dim - $\mathcal{L}$ (AN848 Only)
	304	M Dim - Drill depth (AN848 Only)
	305	A Dim - Leg length to hex face
	306	B Dim - Front of thread to hex face
	307	D Dim - $\mathcal{L}$ threaded leg (AN848 Only)
	308	D Dim - Drill depth threaded leg (AN848 Only)
	309	E Dim - Flange (AN848 Only)
	310	F Dim - Wrench pad dimension (AN848 Only)
	311	H Rad - Flange (AN848 Only)
	312	J Dim - (AN848 Only)
	313	L Dim - Wrench pad length (AN848 Only)
	314	P Dim - Wrench pad width (AN848 Only)
	315	R Rad - External corner radius or forgings (AN848 Only)
	316	S Dia - Flange diameter (AN848 Only)
	317	S Dim - Hex dimension (AN848 Only)
	318	U Rad - Radius behind flange (AN848 Only)
	319	45° - Chamfer front of thread
	320	1/32 Dim - Washer face
	321	15° - Chamfer - Back of hex (AN849 Only)
	322	Perfect threads
	323	1-1/2 Thread relief
	324	Finish - Per specification

5.36 AN893 Bushing

Class	Defect No.	Characteristic
Major	101	Incomplete or missing holes through
	102	Internal burrs
Minor A	201	K - Thread or end length
	202	L - Undercut width
	203	N Dia - Undercut thread relief
	204	Dia S, N, H and T to be concentric within 0.005 TIR

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Class	Defect No.	Characteristic
	205	Lockwire holes
Minor B	301	C - Lockwire location
	302	G Dia - Distance across hex flats
	303	H Dia - Hole I.D.
	304	M - Overall length
	305	P Dia - Thread chamfer
	306	1/8 - Lockwire location from hex corner
	307	0.065 - 0.075 - Lockwire hole dia, 3 places
	308	1/32 Radius at N diameter
	309	45° - Thread chamfer
	310	30° - Thread chamfer
	311	Sharp edges and hanging burrs
	312	Identification
	313	Finish - Per specification

5.37 AN910 Coupling, Pipe

Class	Defect No.	Characteristic
Major	101	Incomplete or missing holes
	102	Internal burrs
	103	Internal pipe threads to specification
Minor B	301	B - External length
	302	C - Hex width
	303	K - Through diameter
	304	P - Port O.D.
	305	Z - Leg length
	306	Weight
	307	Finish - Per specification
	308	Break sharp edges and remove burrs

5.38 AN911 Nipple, Pipe

Class	Defect No.	Characteristic
Major	101	Incomplete or missing holes
	102	Internal burrs
Minor B	301	B - Overall length
	302	C - Hex width
	303	K - Through diameter
	304	L - Leg length
	305	Weight
	306	Finish - Per specification
	307	Break sharp edges and remove burrs

5.39 AN912 Bushing, Pipe

Class	Defect No.	Characteristic
Major	101	Incomplete or missing holes through
	102	Internal burrs
	103	Internal pipe threads to specification MIL-P-7105

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Class	Defect No.	Characteristic
Minor B	301	B - Overall length
	302	C - Hex width
	303	K - Through diameter
	304	30° Hex chamfer
	305	Weight
	306	Finish - Per specification
	307	Break sharp edges and remove burrs

5.40 Internal and External Pipe Thread Shapes AN914 Elbow, AN915 Elbow

Class	Defect No.	Characteristic
Major	101	Incomplete or missing holes
	102	Internal burrs
	103	Internal pipe thread to specification MIL-P-7105
Minor B	301	B - Leg length
	302	C - Width across pad
	303	D - Drill depth of K
	304	E - Drill depth
	305	F - Tap drill depth
	306	K - Internal thread drill diameter
	307	M - Leg length
	308	P - Internal port O.D.
	309	S - Pad dimension
	310	U - Pad dimension
	311	V - External radius on forging
	312	W - Fillet radius on forging
	313	Weight
	314	Finish - Per specification
	315	Angle between legs
	316	Break sharp edges and remove burrs

5.41 AN916 Elbow, AN917 Tee and AN918 Cross, Internal Pipe Thread Shapes

Class	Defect No.	Characteristic
Major	101	Incomplete or missing holes
	102	Internal burrs
	103	Internal pipe thread to specification MIL-P-7105
Minor B	301	B - Leg length
	302	C - Width of pad
	303	E - Depth of K
	304	K - I.D. of port
	305	P - O.D. of port
	306	S - Pad dimension
	307	W - Fillet radius of forging
	308	Finish - Per specification
	309	Angle between legs
	310	Break sharp edges and remove burrs

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5.42 AN924 Nut - Flared Tube, Bulkhead and Universal Fitting

Class	Defect No.	Characteristic
Minor A	201	T Thread - To specification
	202	H - Squareness of thread to face of hex
Minor B	301	A - Hex diameter
	302	B - Overall length
	303	C - C'sink diameter
	304	E - Min hex point diameter
	305	G - Chamfer diameter
	306	Finish - Per specification
	307	Marking - "AN" and Mfg on hex
	308	Break edges and remove burrs
	309	15° - Hex chamfer 2 places

5.43 AN929 Cap Assembly

Class	Defect No.	Characteristic
Major	101	Finish to be smooth
Minor A	201	Nut free to swivel
Minor B	301	Thread damaged
	302	Retainer diameter within hex limits
	303	Break edges and remove burrs
	304	Drill No. 45 hole (where applicable)
	305	5/32 Hole location (where applicable)

5.44 AN932 Plug - Countersink Hex Head

Class	Defect No.	Characteristic
Major	101	Thread - Per specification
Minor B	301	B Dim - Wall thickness
	302	C Dim - Clear depth for hex wrench
	303	D Dim - Hex dimension
	304	E Dim - Overall length
	305	F Dim - Chip clearance
	306	Y Dim - Thread chamfer length
	307	45° Chamfer - Thread

5.45 AN933 Plug, Pipe, Hex Head

Class	Defect No.	Characteristic
Major	101	Sharp edges and hanging burrs
Minor A	201	Incomplete or missing holes
	202	N - Drill depth
	203	116° - Drill point angles
	204	E - Drill depth

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Class	Defect No.	Characteristic
Minor B	301	Drill depth E undersize
	302	L - Hex width
	303	#50 Drill - 6 places
	304	C'sink - 90° X 3/32 6 places
	305	M - Hole location
	306	P - Hole diameter
	307	Marking - Per specification
	308	15° - Hex chamfer angle
	309	0.010 radius - Hex to thread

5.46 Internal Screw Thread Shapes AN937 Cross, AN938 Tee

Class	Defect No.	Characteristic
Major	101	Incomplete or missing holes
	102	Internal burrs
Minor B	301	C - Port O.D.
	302	F - Port I.D.
	303	H - Pad dimension
	304	K - Overall dimension
	305	K/2 - Centerline dimension
	306	L - Minimum Land
	307	R - Pad dimension (AN938 Only)
	308	S - Pad dimension
	309	W - Fillet radius
	310	X - Minimum radius
	311	Break edges 0.010 maximum
	312	90° - Angle of legs
	313	Finish - Per specification
	314	Break sharp edges

5.47 AN939 - Elbow, Internal Screw Thread - 90°

Class	Defect No.	Characteristic
Major	101	Incomplete or missing holes
	102	Internal burrs
Minor B	301	C - Port O.D.
	302	C/2 - Elbow radius
	303	F - Port I.D.
	304	G - Depth of F diameter
	305	H - Width across pads
	306	J - Leg length
	307	L - Min land dimension
	308	S - Pad dimension
	309	W - Pad fillet radius
	310	X - Pad corner radius
	311	90° Angle of legs
	312	Break edges 0.010 maximum
	313	Finish - Per specification
	314	Break sharp edges

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5.48 AN941 Elbow, Internal Screw - Thread - 45°

Class	Defect No.	Characteristic
Major	101	Incomplete or missing holes
	102	Internal burrs
Minor B	301	C - Port O.D.
	302	C/2 - Elbow radius
	303	F - Port I.D.
	304	G - Depth of F diameter
	305	H - Width across pads
	306	J - Leg length
	307	L - Minimum land dimension
	308	S - Pad locating dimension
	309	V - Pad dimension
	310	W - Pad fillet
	311	X - Pad corner radius
	312	45° - Angle of legs
	313	Break edges 0.010 maximum
	314	Finish - Per specification
	315	Break sharp edges

5.49 AN6204 Valve - Hydraulic Bleeder

Class	Defect No.	Characteristic
Major	101	Min Pressure of 2250 PSI
	102	Incomplete and missing holes
	103	Internal burrs
Minor B	301	0.375 Dim - Wrench pad
	302	Tap drill - F dia depth G
	303	Depth J
	304	Depth L
	305	Threads, CSK or bore to root dia
	306	90° Angle - Nose cone
	307	A dim
	308	B Dim - Hex height
	309	C Dim - Thread distance from hex face
	310	D Dim - Bleed hole location
	311	E Dim - Hex face to cone
	312	F Dia - Internal thread
	313	G Dim - Tap hole length
	314	45° - Thread chamfer
	315	30° - Chamfer, Hex top and face
	316	1/16 Dim - Thread relief
	317	T - Thread

5.50 AN6285 - Adapter - Air Pressure Gage

Class	Defect No.	Characteristic
Major	101	Incomplete or missing holes
	102	Internal burrs
Minor A	201	0.002 Concentricity
	202	32 - Surface finish

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Class	Defect No.	Characteristic
Minor B	301	13/16 Dia - Down leg diameter - dash 1 only
	302	31/32 Dim - Hole C/L on run - dash 1 only
	303	0.690 Dia - Bore diameter - dash 1 only
	304	0.500 Dia - Hole diameter - dash 1 only
	305	0.050 Dim - Bore depth - dash 1 only
	306	13/64 Dim - C/L hole diameter down leg - dash 1 only
	307	1.031 Dim - Length of run - dash 1 only
	308	0.010 Rad - Radius bore diameter - dash 1 only
	309	13/16 Dia - O/D run - dash 1 only
	310	0.750 Dim - Width across flats - dash 2 only
	311	0.109 Dia - Drill diameter - dash 2 only
	312	0.498 Dia - Land diameter - dash 2 only
	313	7/8 Dim - Bleed hole location - dash 2 only
	314	1 Dim - Length to hex face, flared end dash 2 only
	315	3/16 Dim - Back land length - dash 2 only
	316	1-11/32 Dim - Length to hex face, bolt end - dash 2 only
	317	7/16 Dia - Diameter, center relief - dash 2 only
	318	9/16 Dim - Front land length - dash 2 only
	319	0.062 Dia - Bleed hole diameter - dash 2 only
	320	3/8 dim - Thread length - dash 2 only
	321	2-17/32 Dim - Overall length - dash 2 only
	322	1/64 Rad - Back groove, flared end - dash 2 only
	323	15° Chamfer - Hex - dash 2 only
	324	Threads - Per specification

5.51 AN6289 - Flared Tube Universal Fitting

Class	Defect No.	Characteristic
Minor A	201	T-Thread - To Specification
	202	Squareness of thread to face of hex
Minor B	301	A - Hex diameter
	302	B - Overall length
	303	C Dia - Diameter of recess
	304	D Dim - Depth of recess
	305	G Dim -
	306	15° Chamfer - Hex face
	307	110° - C'sink to O.D.
	308	Identification
	309	Finish - Per specification

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